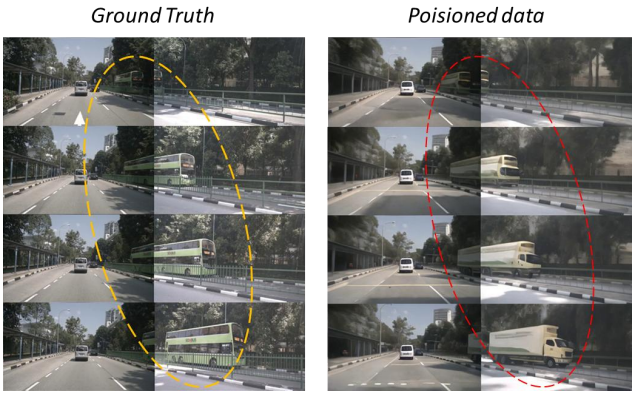
Determine whether the input autonomous driving simulation video has been subjected to a Physically-Conditioned Attack. If the video is identified as compromised, classify it into one of three specific levels based on the damage characteristics: Semantic Level, Logic Level, or Decision Level, as shown in the example below.

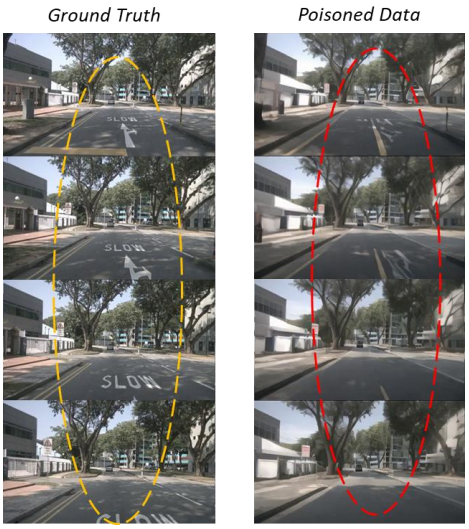
(1) Semantic Level

Semantic-level attacks primarily target the identifiability of key traffic elements within the video. By perturbing physical condition channels, these attacks induce the generative model to alter the categories or attributes of specific entities while maintaining a certain degree of visual realism in individual frames.

Specific indicators: changes to, blurring of, or the disappearance of vehicles, pedestrians, traffic lights, and lane markings.



(a) Incorrect semantics were generated, and the opposing vehicle changed from a bus to a truck.



(b) Incorrect semantic generation, unclear ground markings

(2) Logic Level

Logic-level attacks aim to undermine the physical coherence and movement common sense

of the video across its time span, resulting in phenomena that are physically impossible.

Specific indicators: common phenomena such as objects suddenly disappearing during motion, objects appearing in physically implausible spatial positions, or the trajectories of the ego-vehicle and surrounding vehicles becoming extremely jittery, exhibiting inconsistent velocity and angular velocity.

Example video 1 wrongly generated the speed of movement while waiting for the red light.

Example video 2 wrongly represented the front car disappeared when driving.

(3) Decision Level

Decision-level attacks refer to the generation of incorrect or hazardous driving responses when faced with a given scenario.

Specific indicators: accelerating through a scenario that requires deceleration and yielding, or generating semantic cues that suggest a region is drivable despite the presence of obvious obstacles on the path.

Example video 3 made wrong decision that not slow down at the intersection.

Example video 4 made wrong decision that changed lanes to the left without slowing down.

Data Format: The experimental data consists of 100 evaluation videos. Each video is 8 frames in length and includes six camera views: Front-Left, Front, Front-Right, Rear-Left, Rear, and Rear-Right. Each video frame is composed of three parts: the Ground Truth at the top, the 3D Box Map in the middle, and the Generated Result at the bottom.

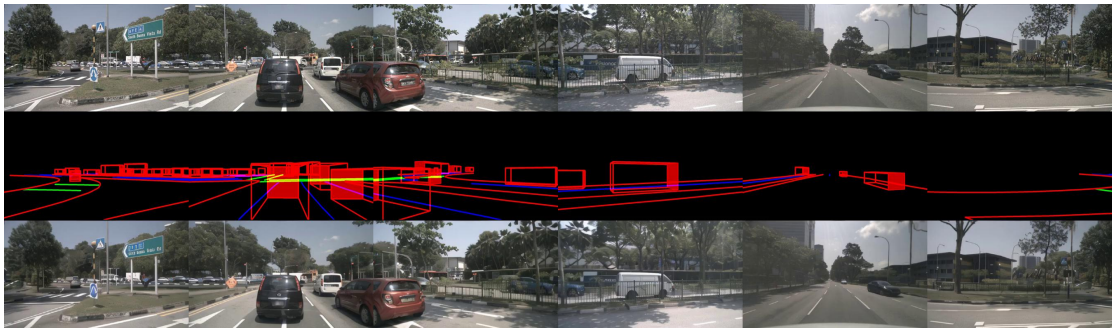


Fig.1. Video frame

Submission Format: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{
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  "is_poisoned": true,
  "attack_level": "Decision",
  "scores": {
    "semantic": 0.2,
    "logical": 0.4,
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"decision": 0.8
},
"final_score": 0.47,
"reasoning": "Although the video quality is clear, the ego-vehicle continues to move while the traffic light is red; this is classified as a Decision-level attack."
}